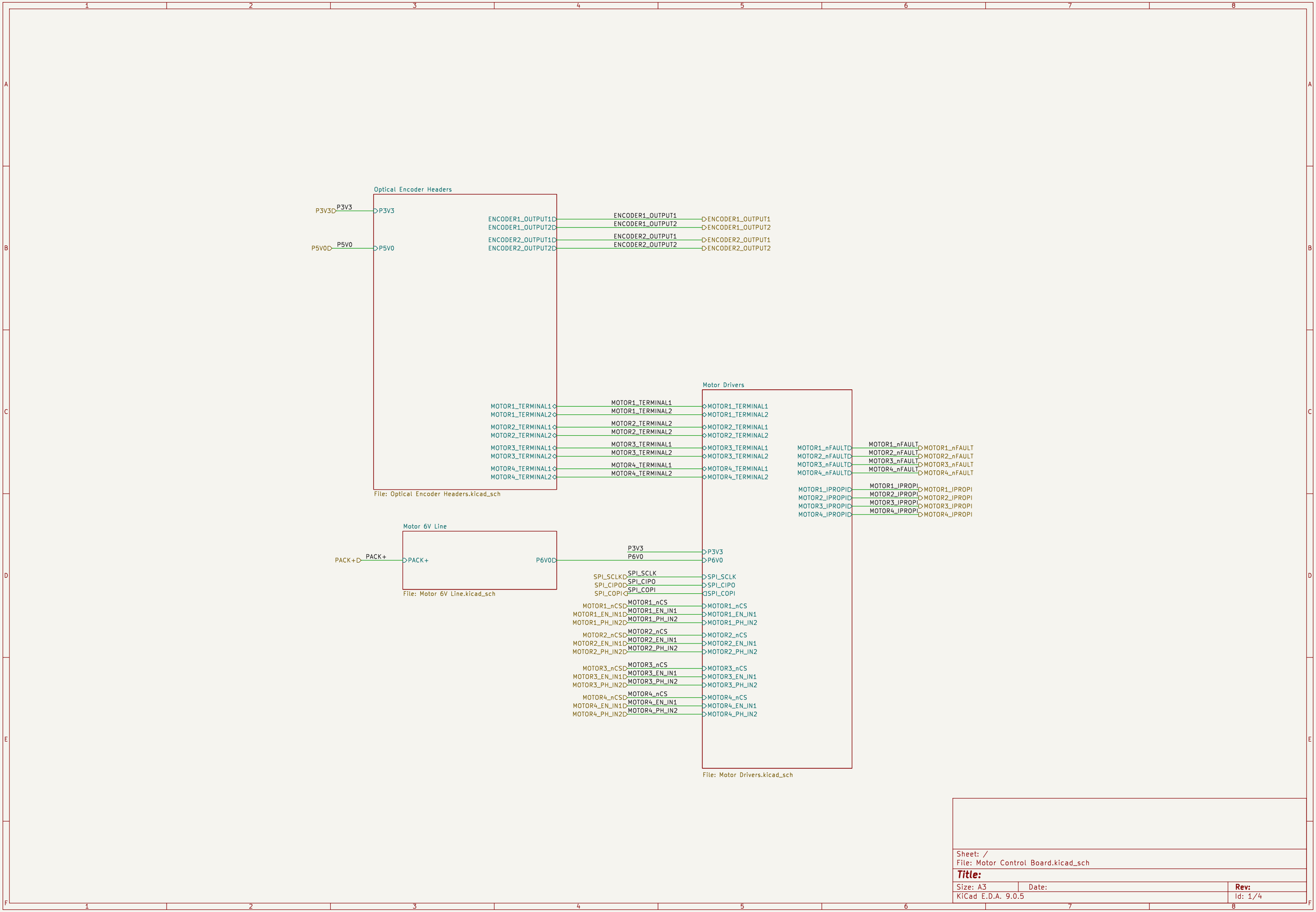
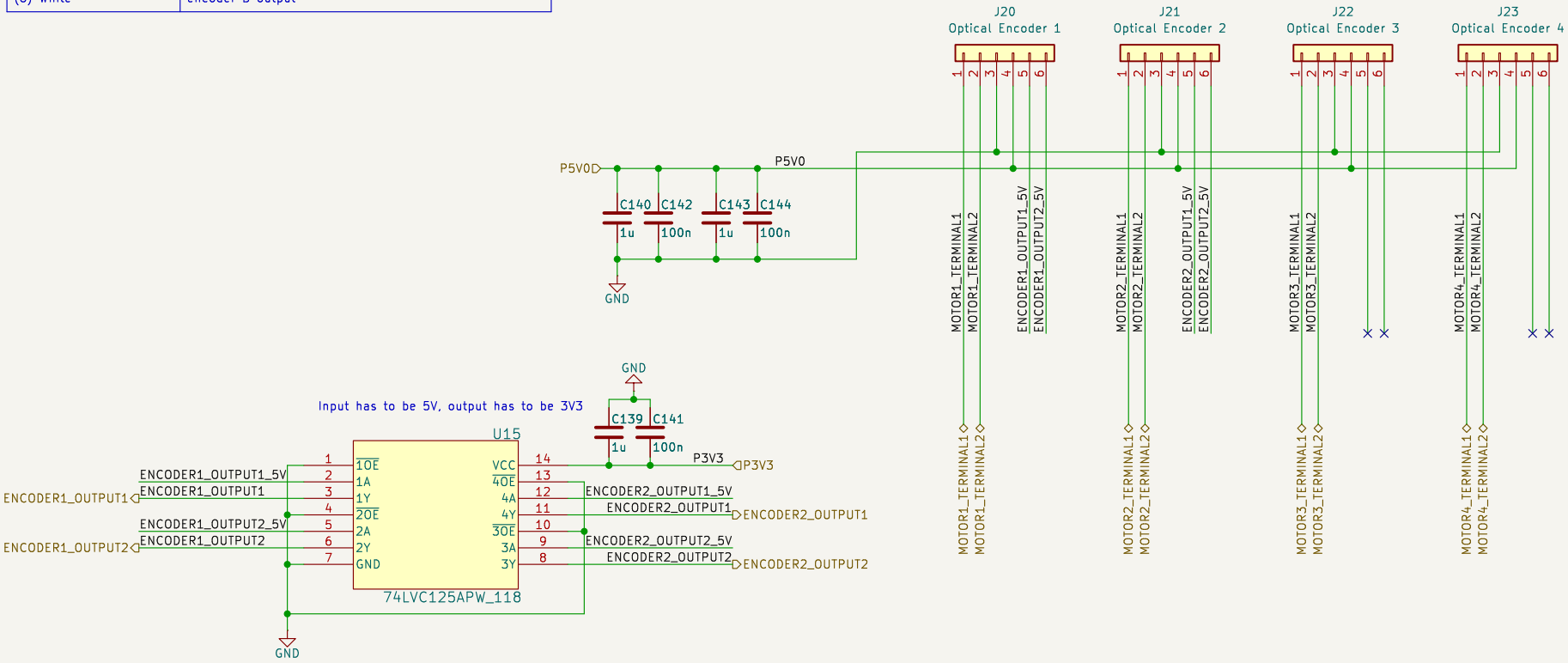


Sheet: /
File: Motor Control Board.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 1/4

(1) Red	motor power (connects to one motor terminal)
(2) Black	motor power (connects to the other motor terminal)
(3) Green	encoder GND
(4) Blue	encoder Vcc (3.5 – 20 V)
(5) Yellow	encoder A output
(6) White	encoder B output



Sheet: /Optical Encoder Headers/
File: Optical Encoder Headers.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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6V/20A
L=440nH
f=1.2MHz
Vin=[9.6,12.6]

D = Vout/Vin = 6/12.6 = 0.476 ; use for worst-case input ripple
I_Ripple = (Vout(1 - Vout/Vin))/(L*f) ; use larger Vin
I_In = Iout * (Vout/Vin) ; use smaller Vin for max current
Ic(RMS,in) = Iout * SQRT[D*(1-D)]
Ic(RMS,out) = I_Ripple / (2*sqrt(3))

I_Ripple = 5.95A
I_In = 12.5A
Ic(RMS,in) = 9.99A -> 12A
Ic(RMS,out) = 1.72A -> 2.1A

1, Input cap bank > 12A rms ripple current
2, Output cap bank > 2.1A rms ripple current

3, Ripple_Vout = I_Ripple * ESR
60mA = 5.95A * ESR, ESR=10.1mOhm

4, 12.5A is the max average current drawn from the source

6V 20A

U14
TPS40305DRCT
VDD
EN/SS
PGOOD
BP
COMP
FB
HDRV
BOOT
SW
LDRV/_OC
EP_1
EP_2
EP_3
EP_4
EP_5

PACK+
C117 10u
C118 10u
C119 10u
C120 10u
C121 10u
C122 10u
C123 10u
C124 10u
C125 1u
C126 3n3
R82 100k
C127 2u2
C128 220p
C129 6n8
R83 1k37
GND

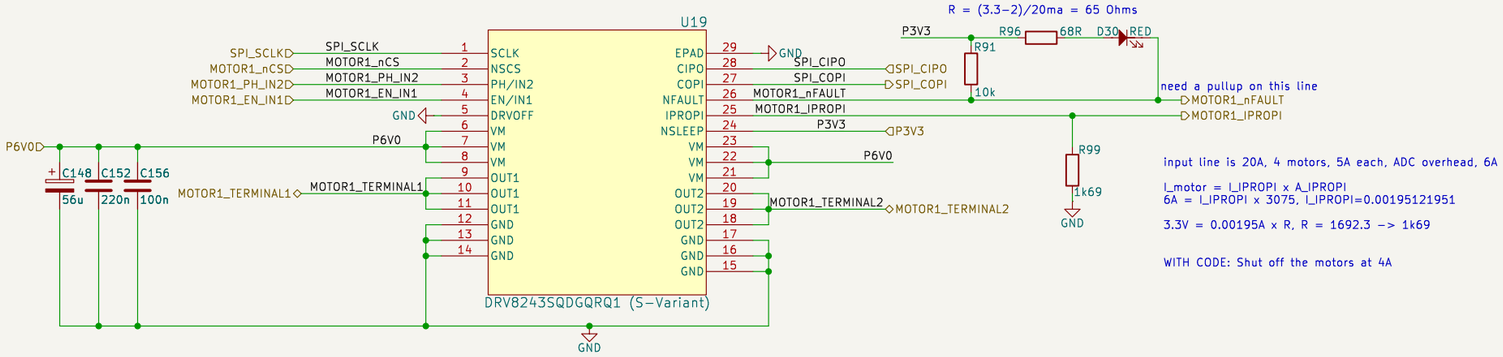
Q13 CSD19531Q5AT
Q14 CSD19531Q5AT
C130 100n
L4 SPB1308-R44M
R84 7k68
R85 309R
R86 10k
R87 1k
C131 1n
C132 1u
C133 10u
C134 10u
C135 10u
C136 10u
C137 10u
C138 10u
6V
P6V0

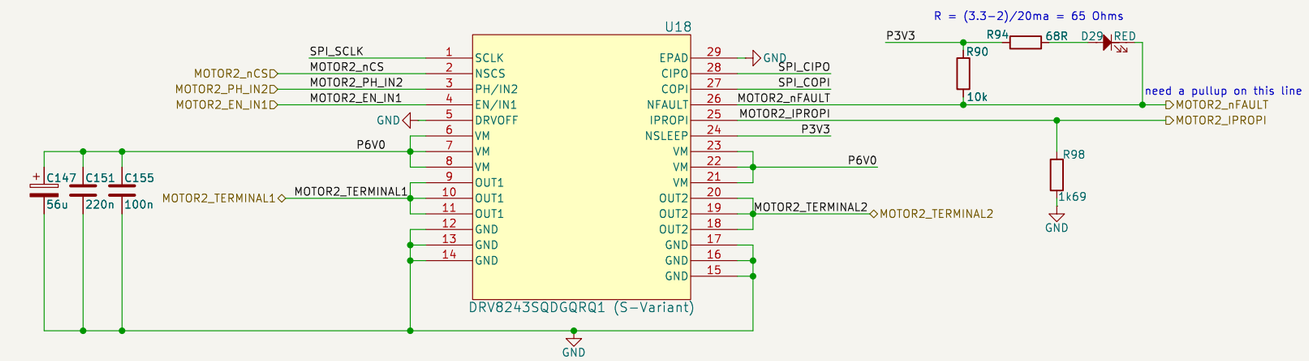
V_OC = (R_OCSET * 2 * I_OCSET_min) + V_OCLOS_min
V_OC = (7.68 kΩ * 2 * 9.5 µA) + (−8 mV)
V_OC = 137.9 mV

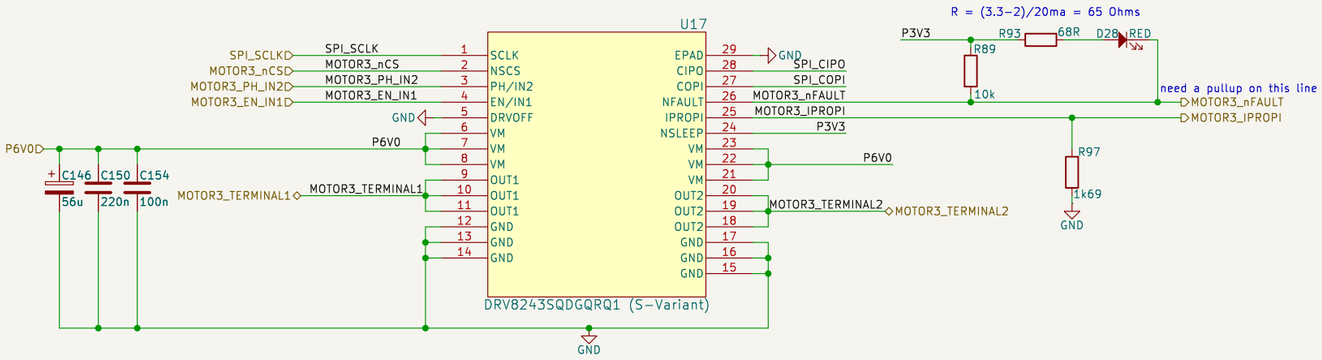
I_valley = V_OC / R_DS(on)
I_valley = 137.9 mV / 7.8 mΩ
I_valley = 17.68 A

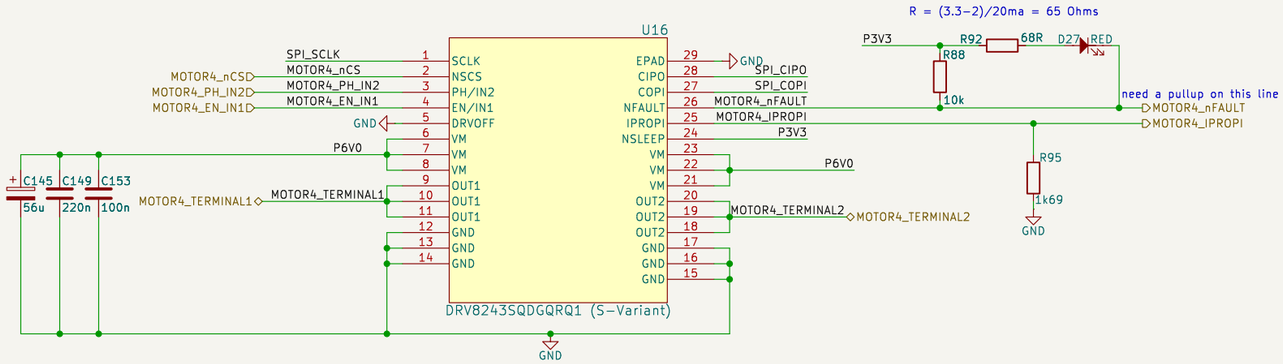
I_SCP = I_valley + (I_RIPPLE / 2)
I_SCP = 17.68 A + (5.95 A / 2)
I_SCP = 20.66 A

Sheet: /Motor 6V Line/
File: Motor 6V Line.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 9.0.5 | Id: 3/4









Sheet: /Motor Drivers/	
File: Motor Drivers.kicad_sch	
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KiCad E.D.A. 9.0.5	**Rev:** Id: 4/4